


Biol 100: Class #9

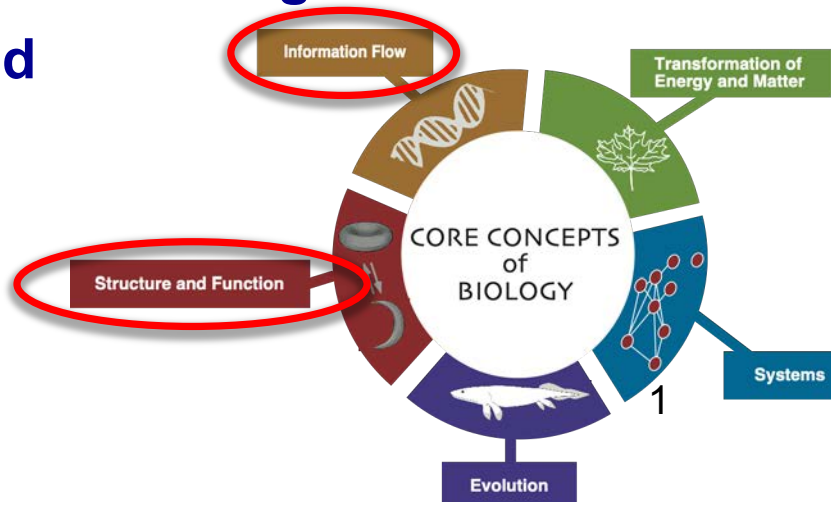
Agenda:

"Formation of Mutations in DNA and Chances of Inheriting a Cancer Mutation"

What is the difference between somatic cells and gametes? What evidence do we have that a cancer may be hereditary? What evidence do we have that a cancer may have been spontaneous? What is the chance of passing on particular traits to your offspring?

How did we do on Exam #1?

- **Welcome, Agenda, & Announcements**
- **Mini-lecture: Fungus, Mutations, and Liver Cancer**
  - **How do mutations happen?**
  - **Global distribution of liver cancer**
- **Key ideas about inheritance of cancer-causing mutations**
  - **Clicker questions: cancer and inheritance**
- **Homework assignments**
- **Return Exam #1**



Sally Pasion, Professor
Kimberly Tanner, Professor

Information Flow



How do we get from eating peanut butter → a mutation?

Aspergillus

Aspergillus is a fungus that grows on nuts and grains in **hot** and **humid** environments.

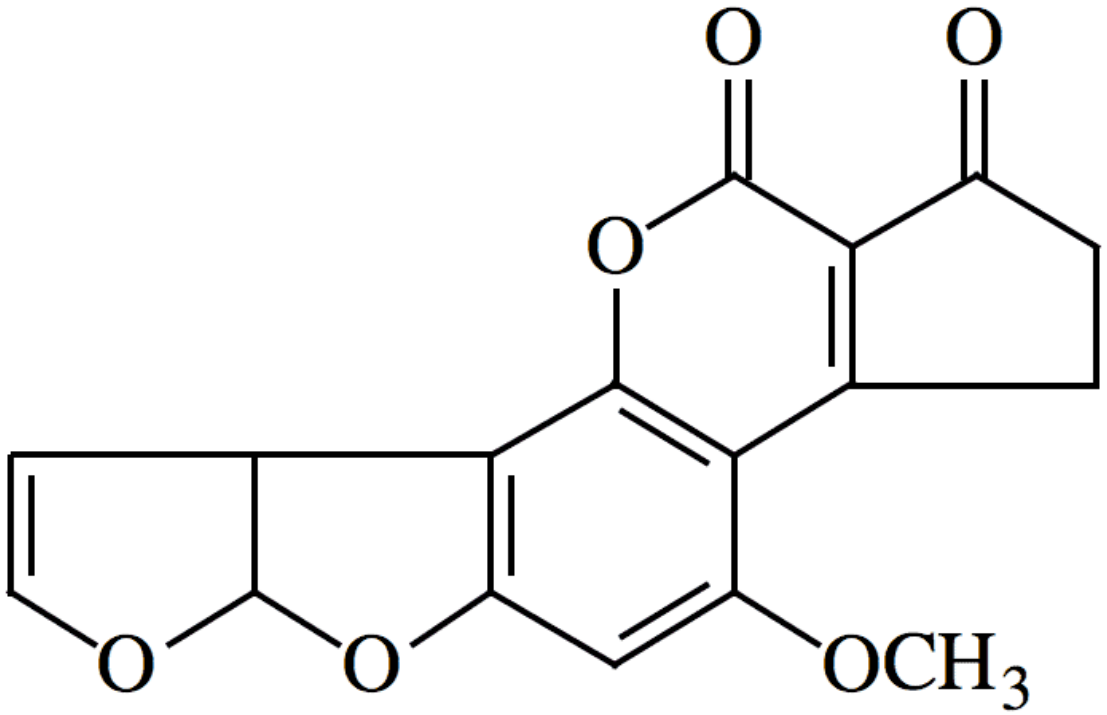




Aspergillus fungi produce aflatoxin



Aspergillus fungi produce **aflatoxin**



It's a chemical. It's a toxin. It's a poison.
It's not a protein. It's not living.



Clicker Question

If **aflatoxin** can cause cancer, it probably does this by directly damaging…

A. Only proteins
B. Only DNA
C. Only RNA
D. Both DNA and RNA
E. DNA, RNA, and Protein



How do mutations actually happen at the level of a DNA molecule in the nucleus of a cell?

How can we figure out the chance of passing a mutation in a cancer-causing gene on to our offspring?



So, what causes mutations?!?!

The world is an imperfect place…

Spontaneous Mutations

(in the normal course of copying DNA, DNA Replication)

Common Mutagens

(things that cause mutations in DNA)

- aflatoxin
- UV (ultraviolet) radiation from sun
- X-Rays
- Variety of chemicals (benzene)
- Cigarette smoke (benzopyrene)
- Inhaled asbestos
- Viruses

…to name a few…

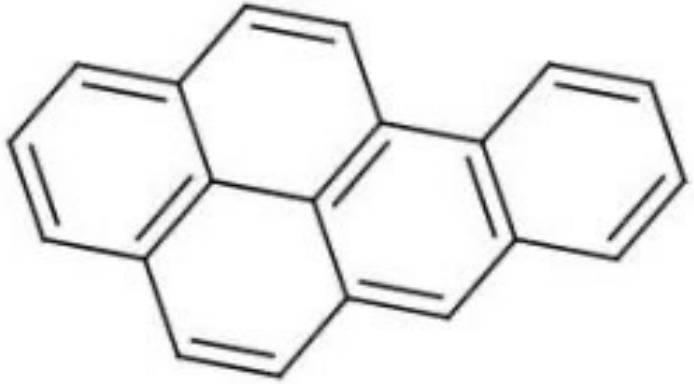
DNA is "mutated" or "has mutations!"

This should not be said about cells…

That said, mutations are important in biology. Think evolution and variation!

From Biol 100, Class #6

Benzopyrene and Aflatoxin are "mutagens"

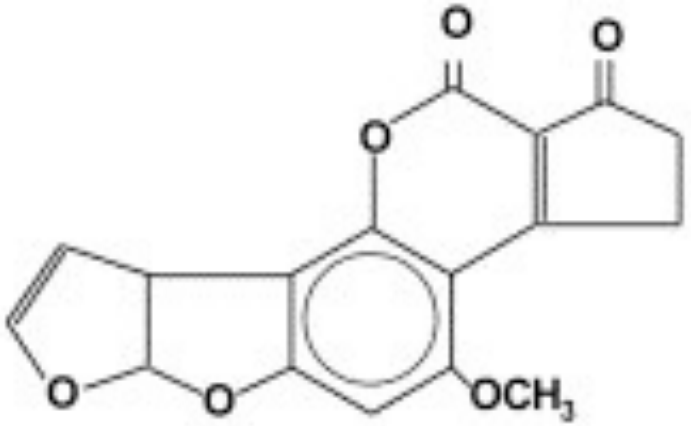


Benzopyrene

Found in cigarette smoke

Aflatoxin

Potential contaminant in peanut butter



Mutagens bind DNA and cause DNA mutations

Aflatoxin

Binds p53 gene and causes mutation



Revisiting Clicker Question

If aflatoxin can cause cancer, it probably does this by directly damaging…

A. Only proteins
B. Only DNA
C. Only RNA
D. Both DNA and RNA
E. DNA, RNA, and Protein

Which regions of the world do you predict have the highest levels of **aflatoxin**?

Which regions of the world do you predict have the highest incidence of liver cancer?

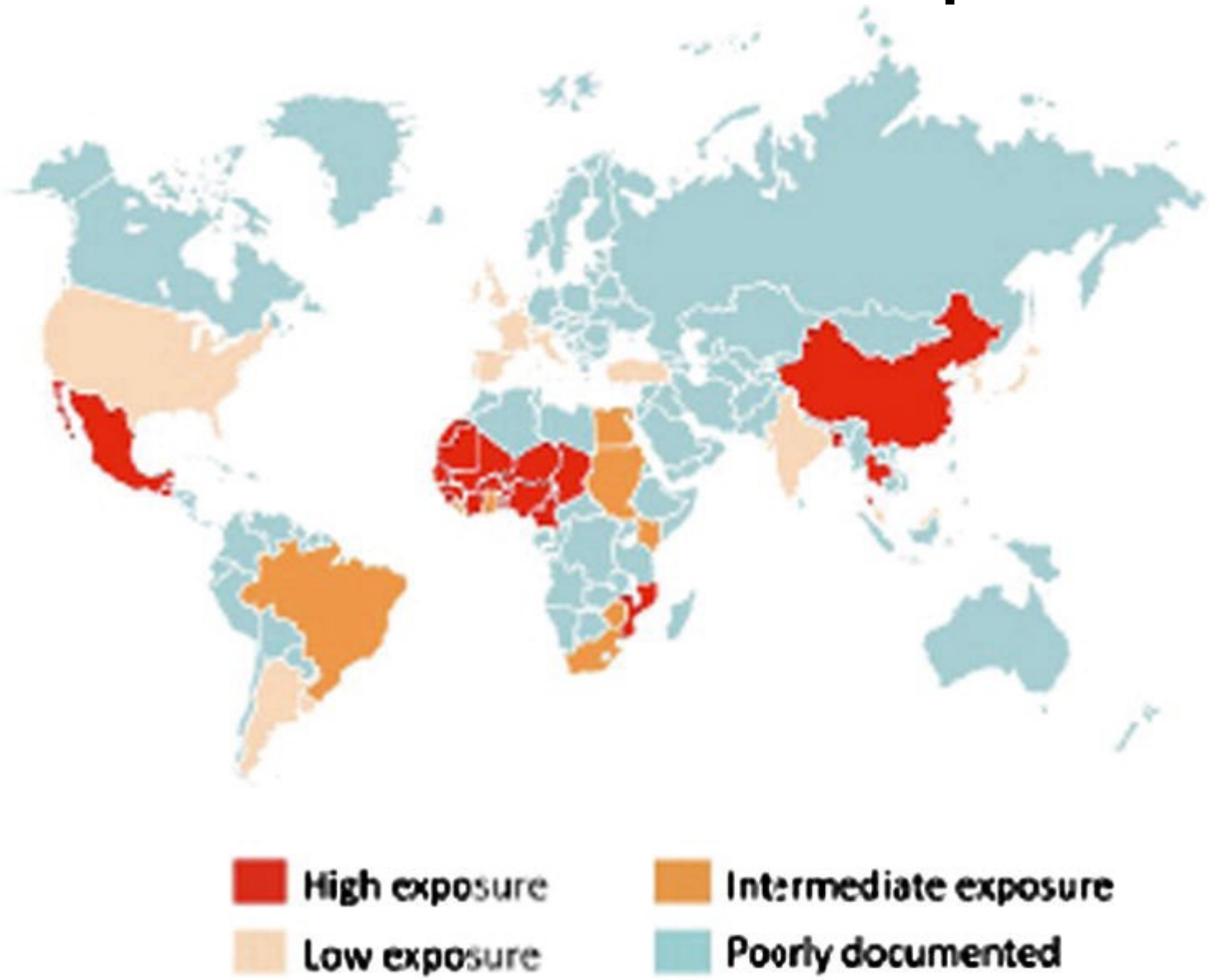


Which regions of the world do you predict have the highest levels of **aflatoxin**?

TABLE DISCUSSION: Where do you predict? Person who traveled the furthest this morning starts.

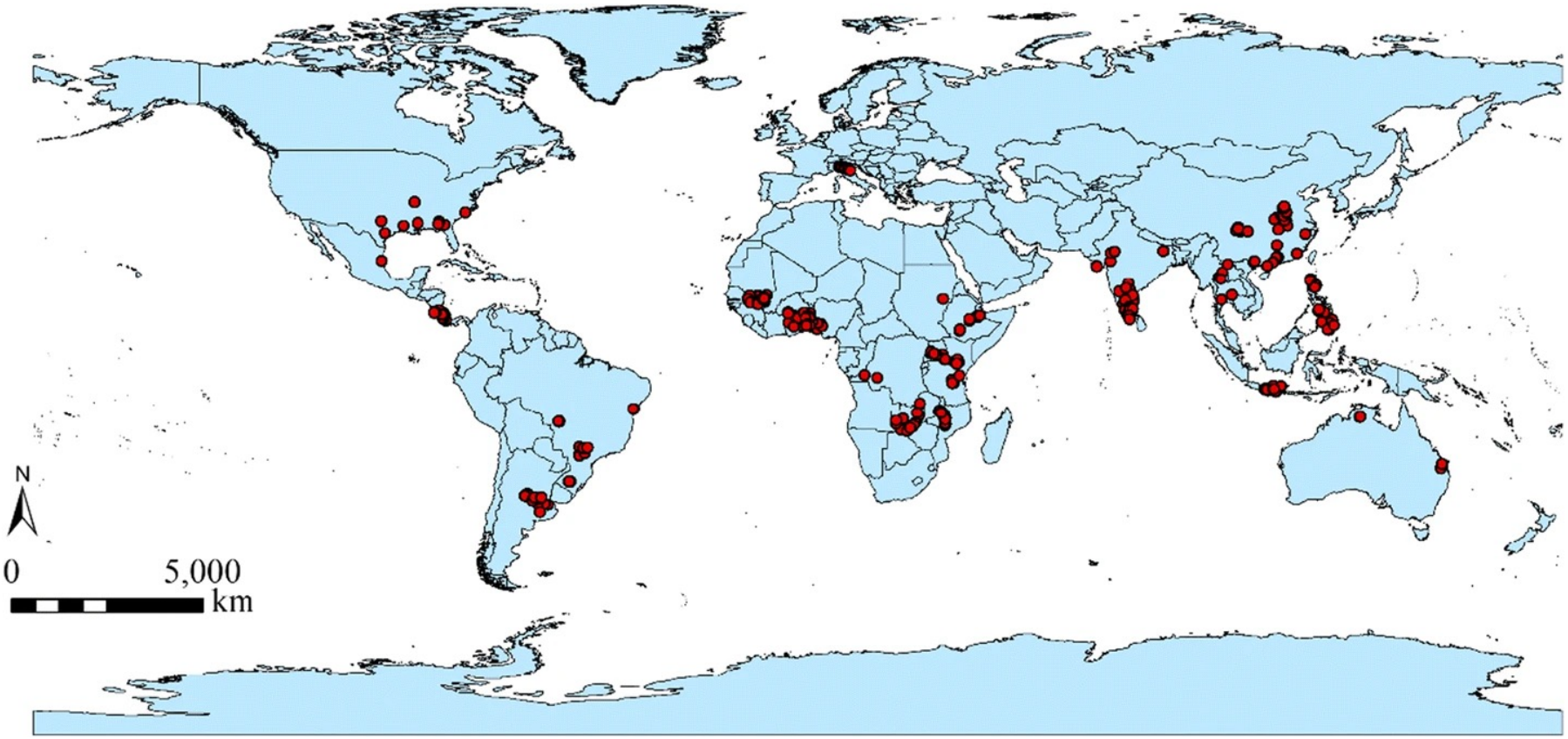


Global aflatoxin exposure



Gouas et al. 2009

Aflatoxin exposure (2020)



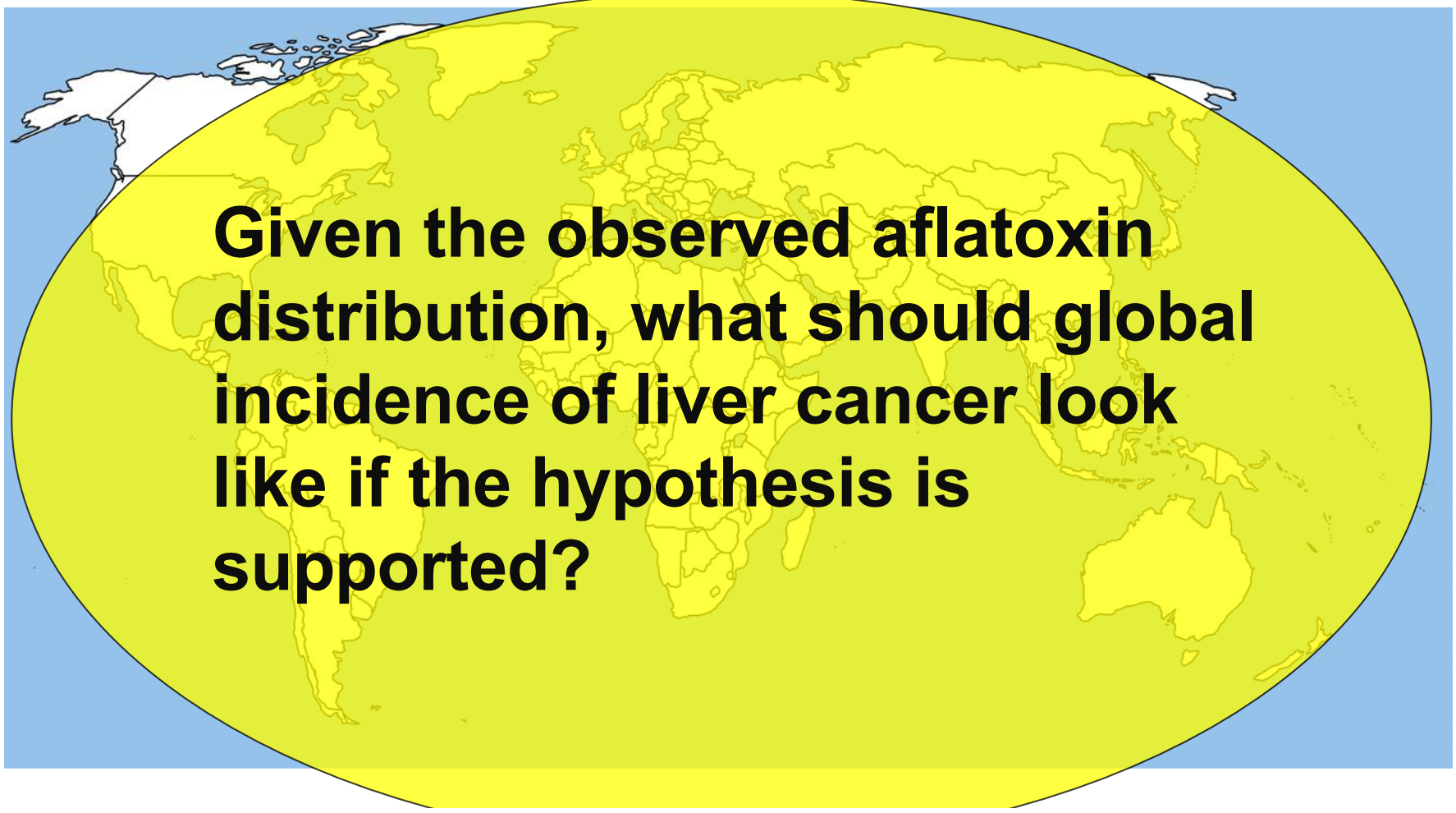
FYI: Here is a more recent aflatoxin exposure map

Haerani et al., 2020

Pair discussion

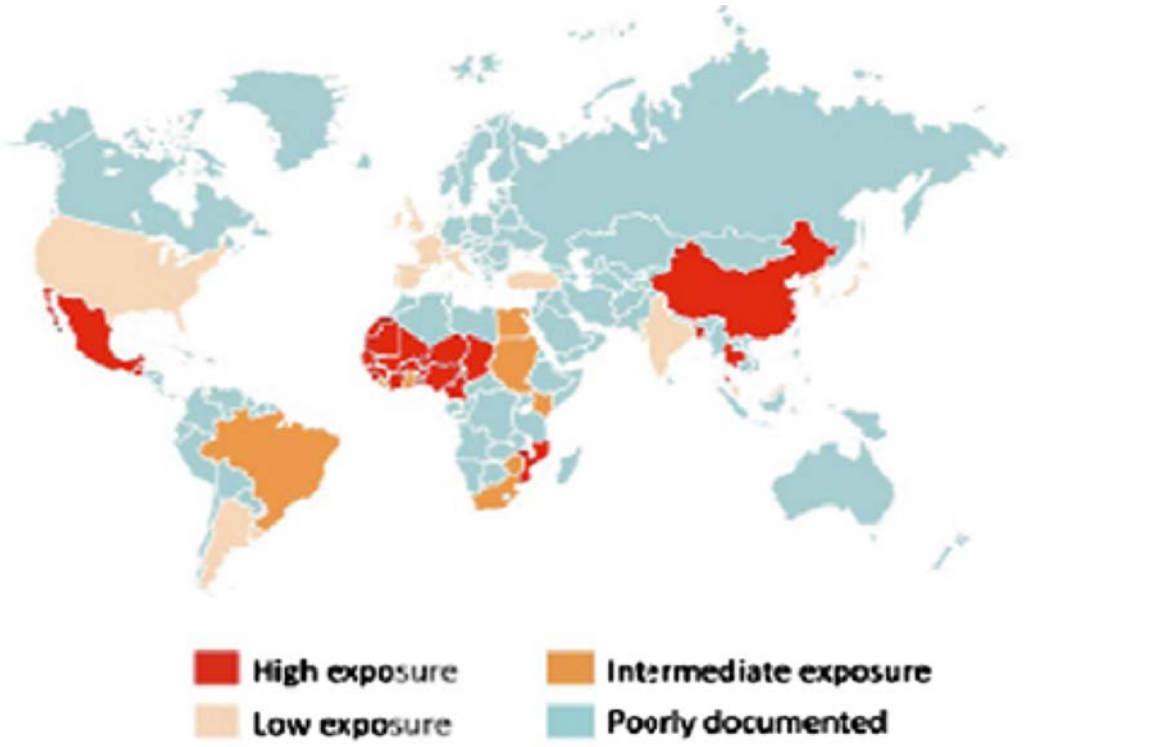
with your shoulder buddy or your team of three!

Person who traveled the shortest distance to class this morning starts.

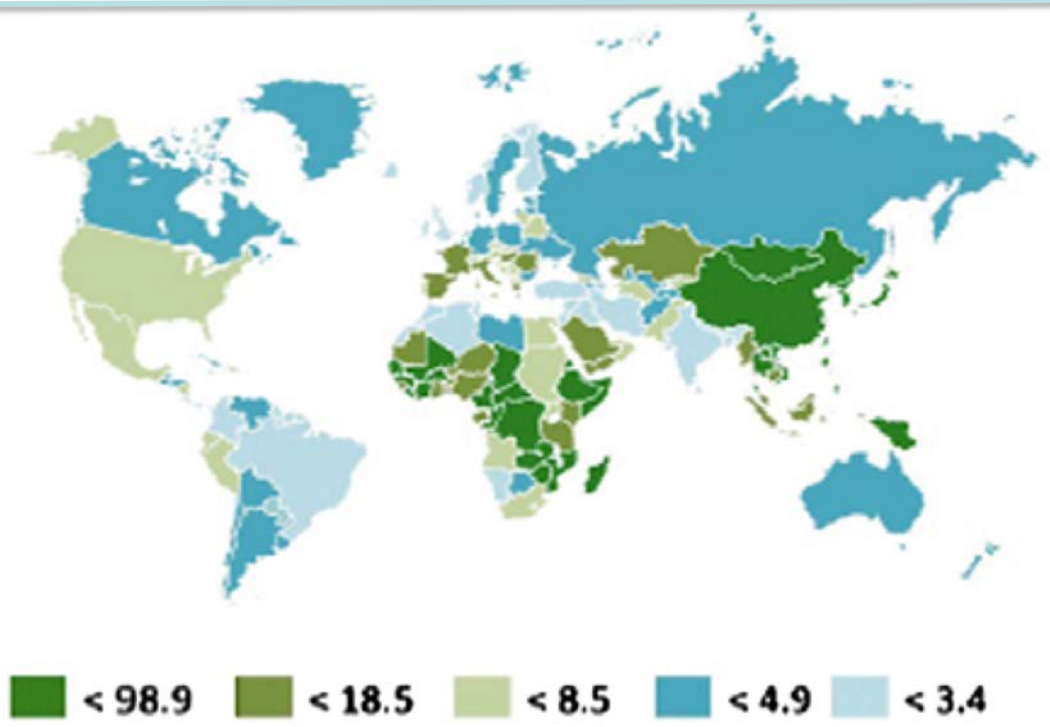


Given the observed aflatoxin distribution, what should global incidence of liver cancer look like if the hypothesis is supported?

Global aflatoxin exposure

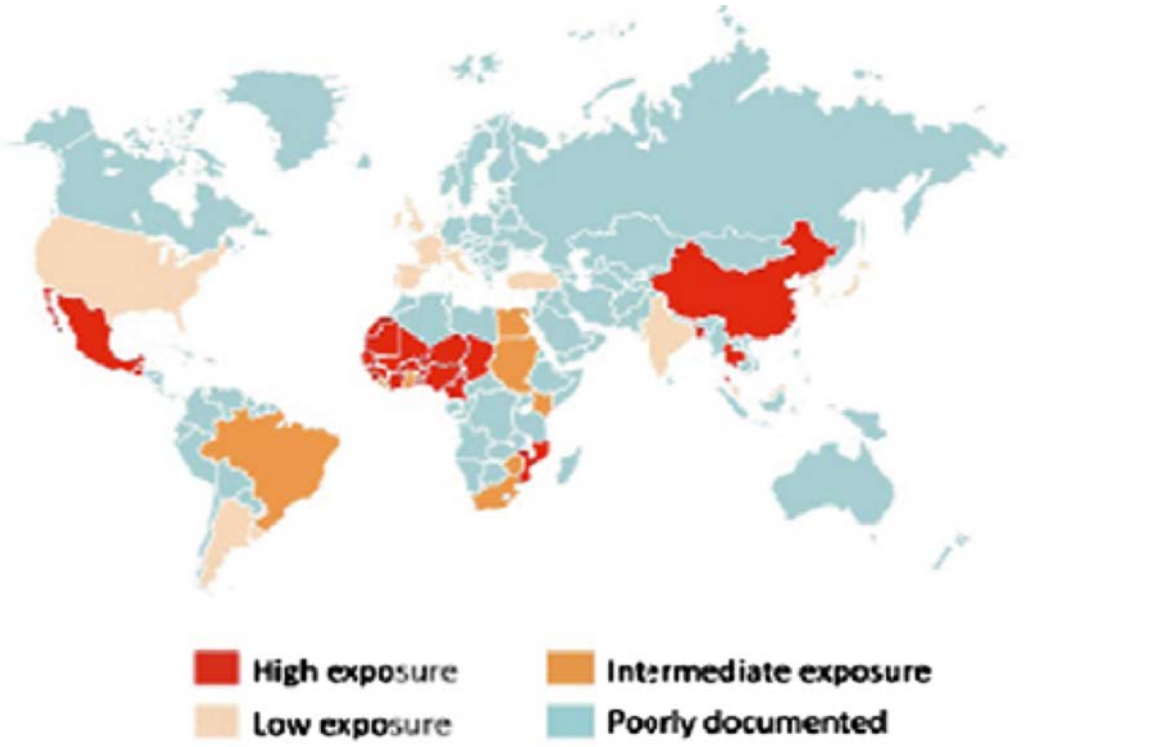


Global incidence of liver cancer

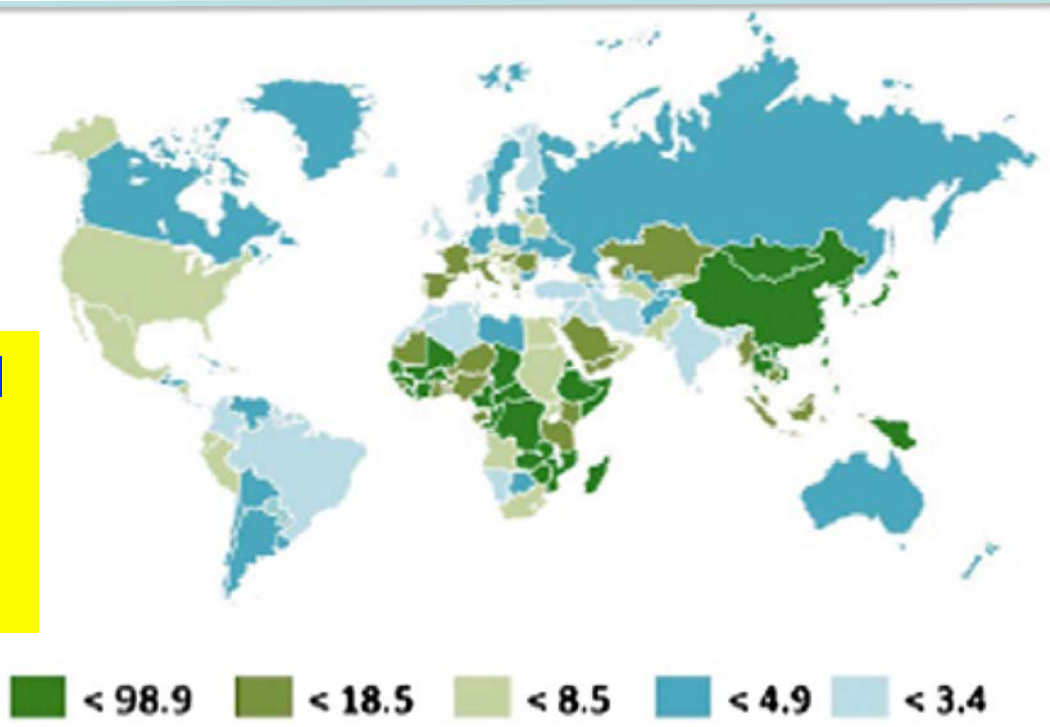


Gouas et al. 2009

Global aflatoxin exposure



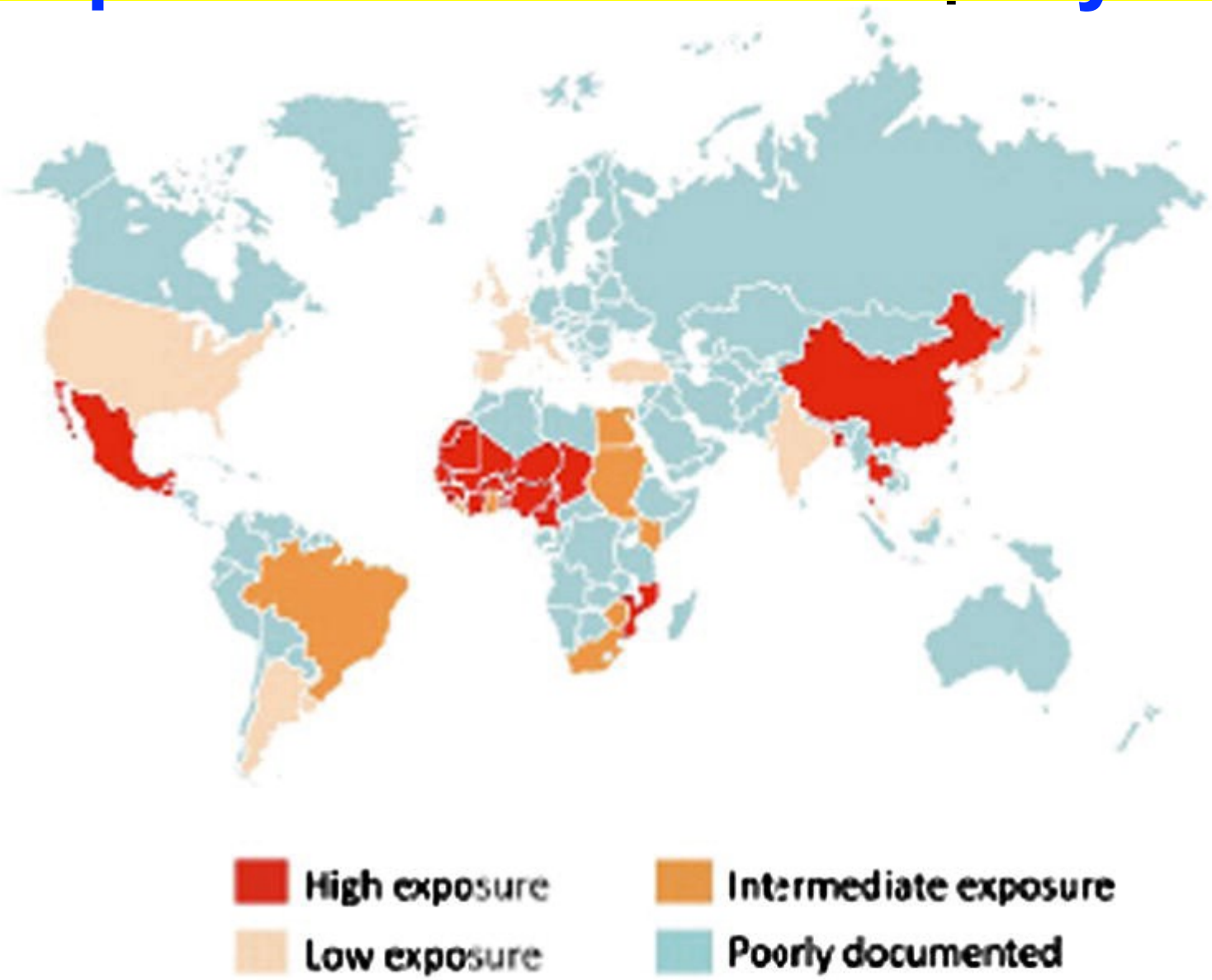
Global incidence of liver cancer



Is the pattern for global incidence of liver cancer consistent with our prediction?

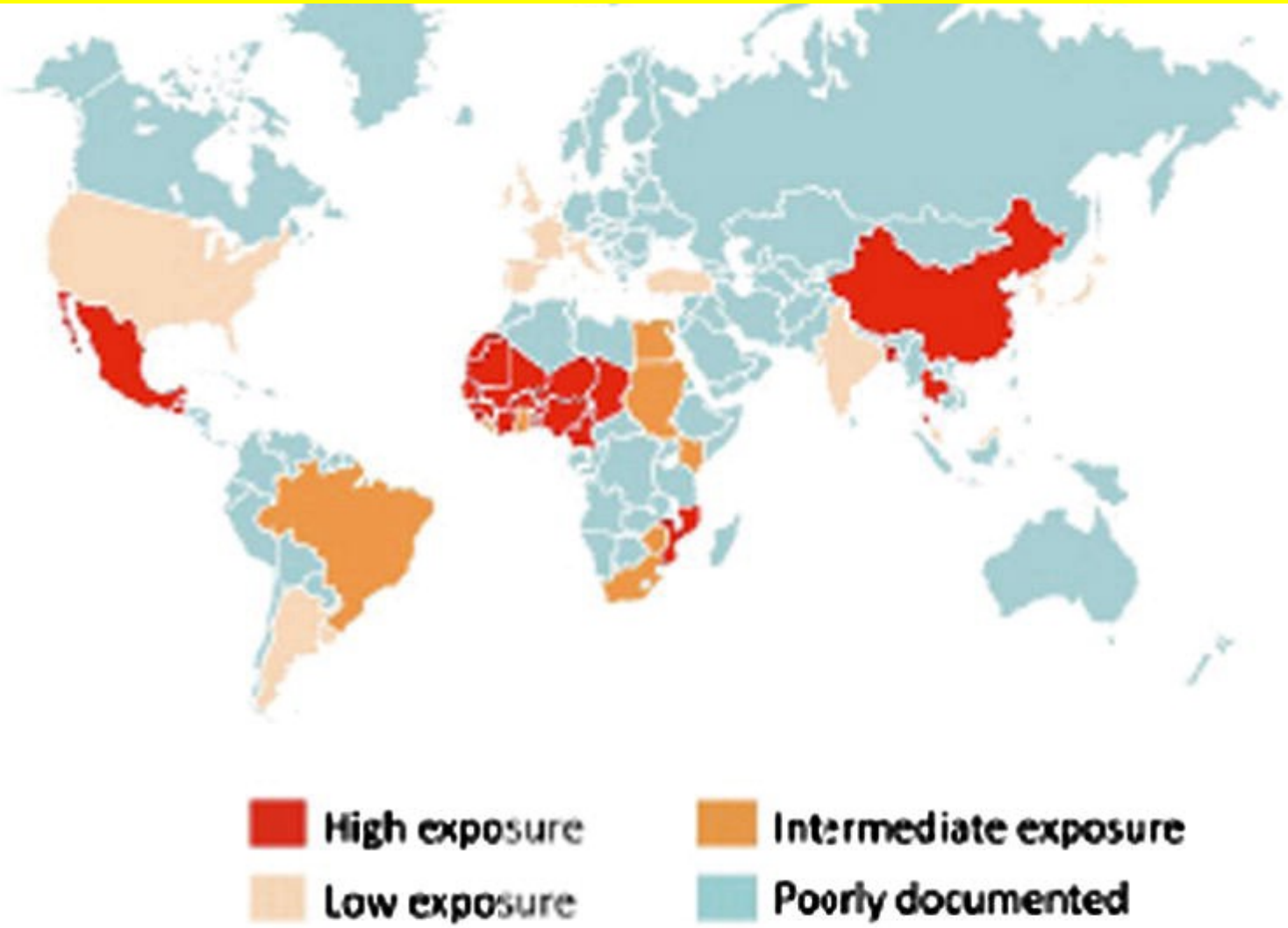
Gouas et al. 2009

Is the biological risk for aflatoxin exposure the same for everyone??



Gouas et al. 2009

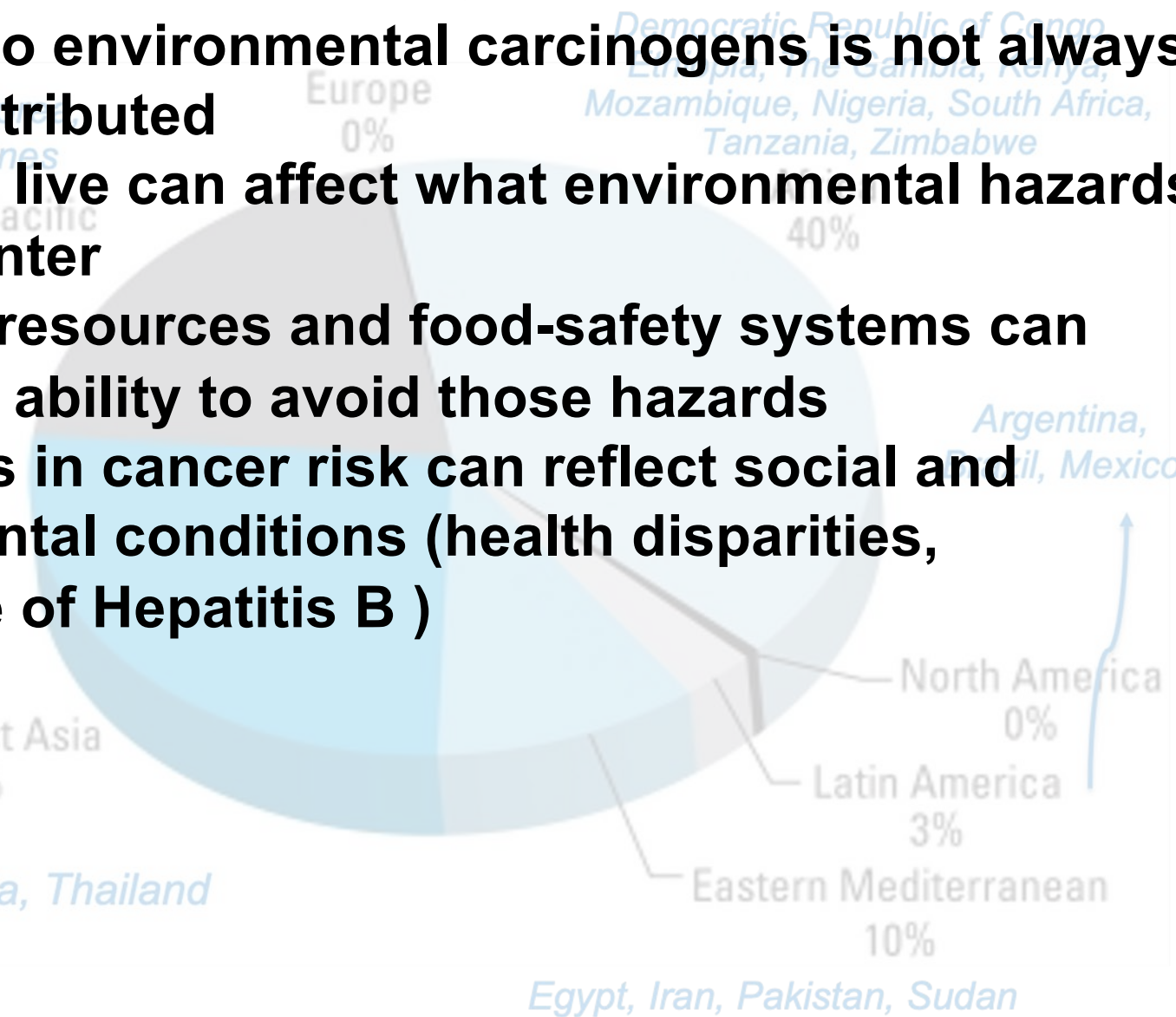
Location! Location! Location!
Why might where someone lives affect their exposure?



Gouas et al. 2009

Aflatoxin Exposure & Social Justice

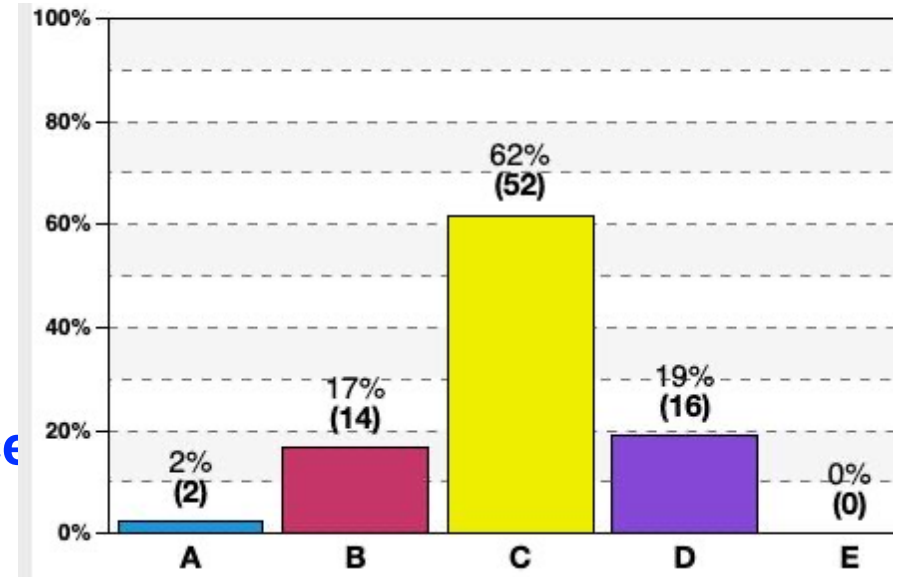
- Exposure to environmental carcinogens is not always equally distributed
- Where you live can affect what environmental hazards you encounter
- Economic resources and food-safety systems can affect your ability to avoid those hazards
- Differences in cancer risk can reflect social and environmental conditions (health disparities, prevalence of Hepatitis B)



Liu and Wu, 2010



Clicker Question from before Exam 1



A young man develops skin cance not spread to any other tissues.

The mutation responsible for the cancer arose in a single cell in his skin.

If he and a female partner (who does not have skin cancer) have children after his skin cancer diagnosis, which of the following is most likely true?

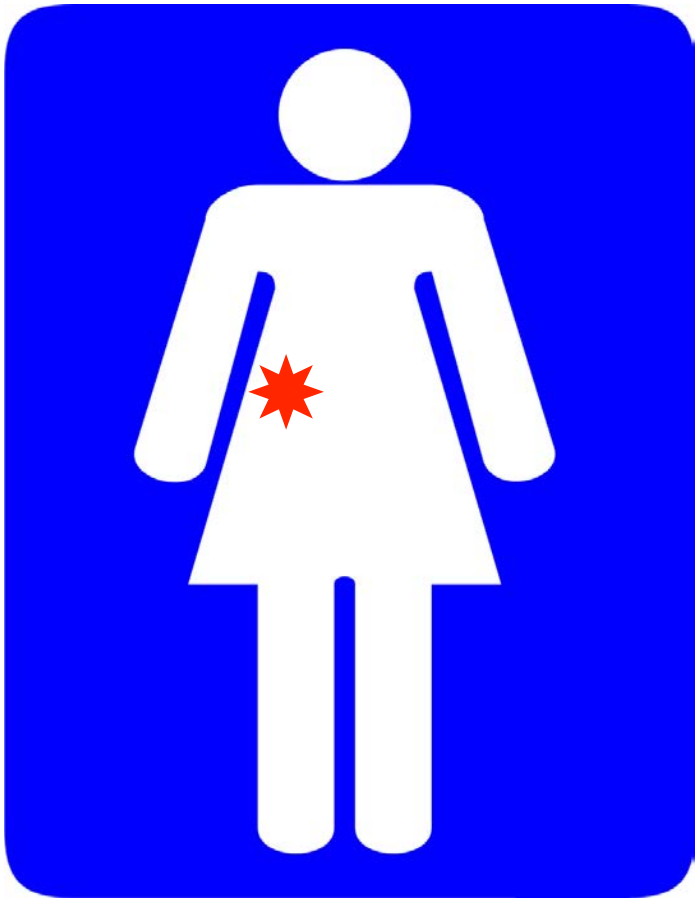
A. All the man's children will inherit the mutation

B. All the man's children will inherit the mutation if the mutation is dominant

C. Some of the man's children may inherit the mutation depending on which of his chromosomes they inherit

D. None of the man's children will inherit the mutation

Clicker Question



A person has developed liver cancer due to aflatoxin exposure. What is the chance that their future biological child will have liver cancer?

A. They will definitely inherit cancer.

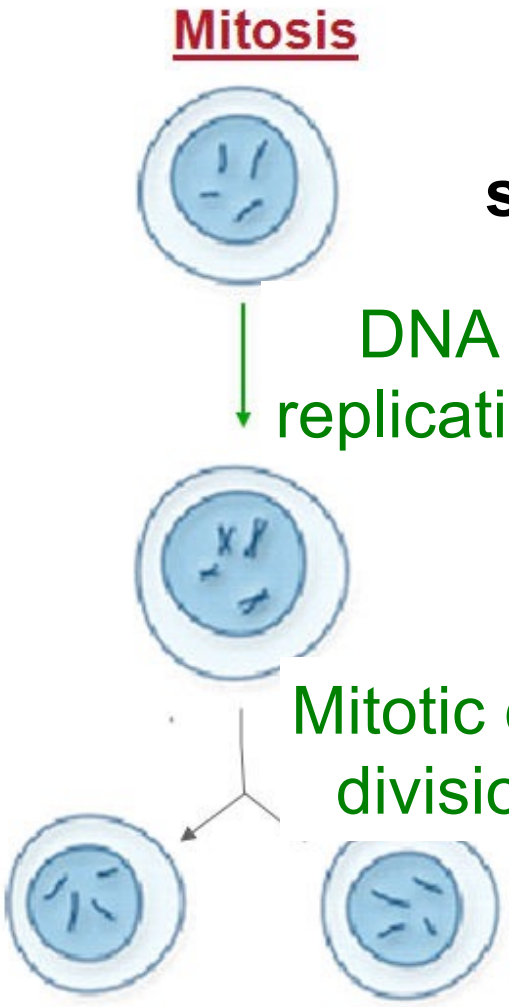
B. They have the normal risk of developing cancer.

C. They will not likely develop cancer by inheritance.

D. Both B and C

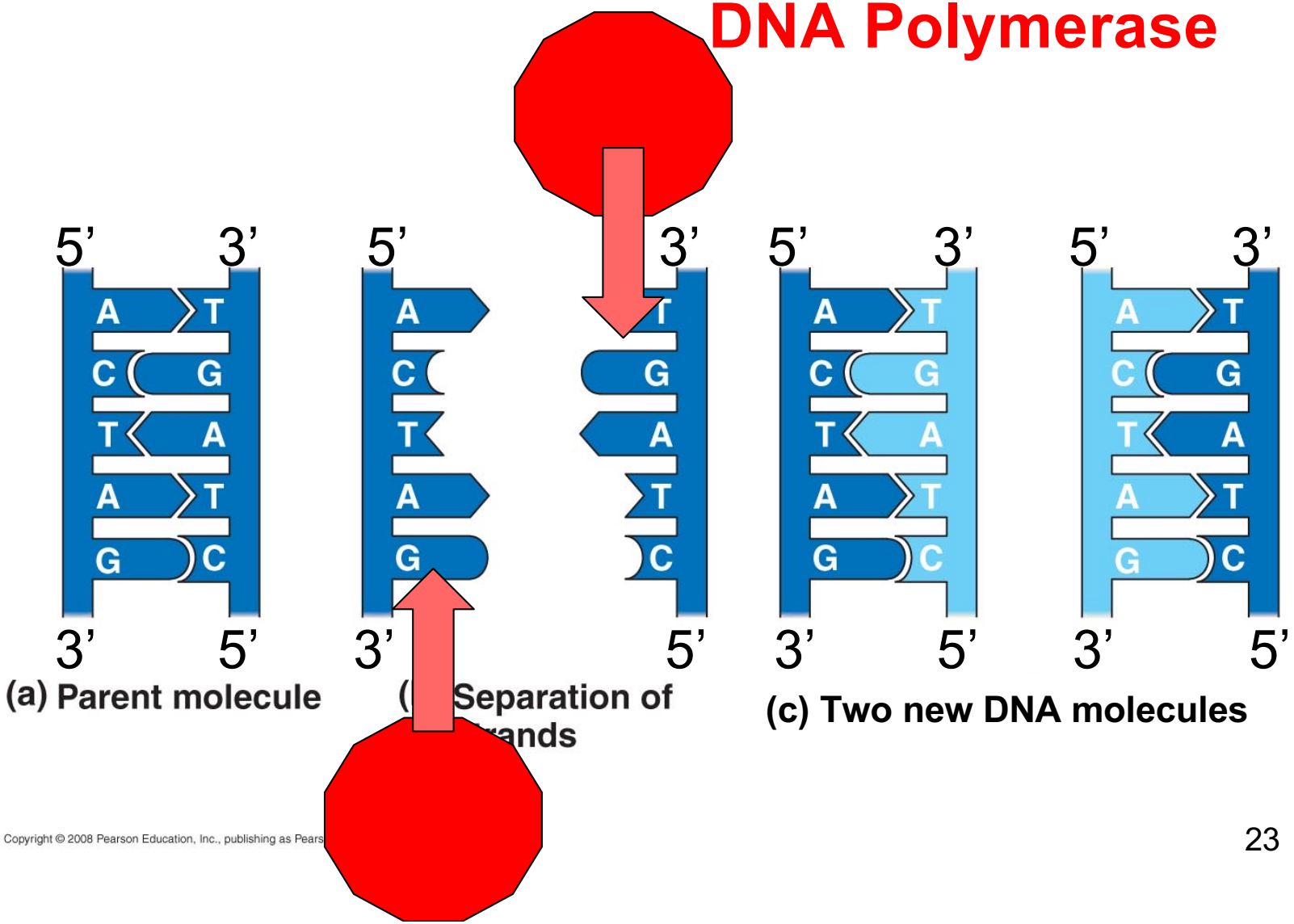
E. I don't know

Key Idea 1: Mitosis occurs in somatic cells and produces two identical cells



Somatic cell =
all cells that are NOT in the germline making eggs/sperm

Big Idea: Mutations can happen during DNA replication when cells are preparing to divide and go through cell division (*mitosis*).



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How do mutations actually happen at the level of a DNA molecule in the nucleus of a cell?

How can we figure out the chance of passing a cancer-causing mutation on to our offspring?



How can we figure out the chance of passing a cancer-causing mutation on to our offspring?

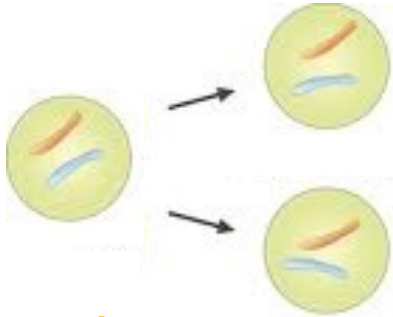
How is the chance DIFFERENT if the mutation is in a DNA molecule in…
a SOMATIC CELL versus
a GAMETE (SEX CELL; SPERM; EGG)?

Where we've been and where we're going…

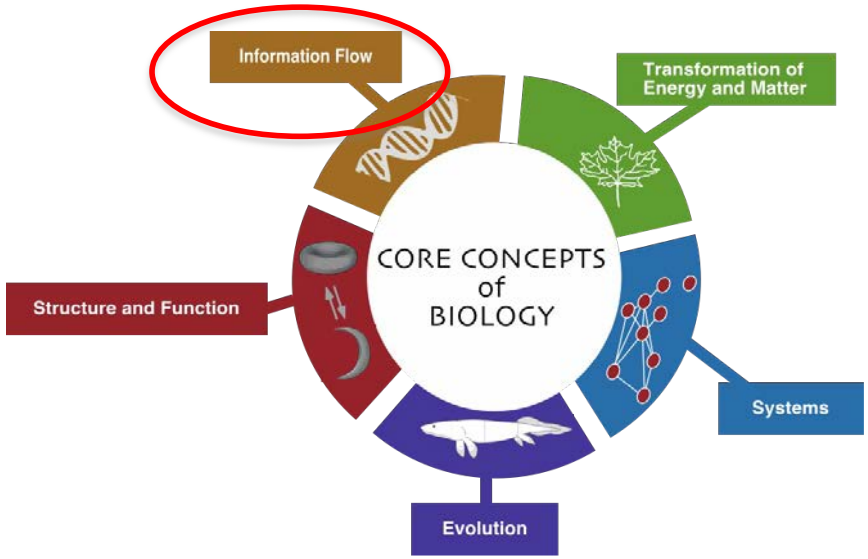
Information Flow

- Gene Expression (Central Dogma)
  - From DNA (gene) through RNA to Protein to Trait
- Cell Division (mitosis)
  - From a parent cell to two new cells

two new cells



- Inheritance (meiosis)
  - From one generation to the next generation



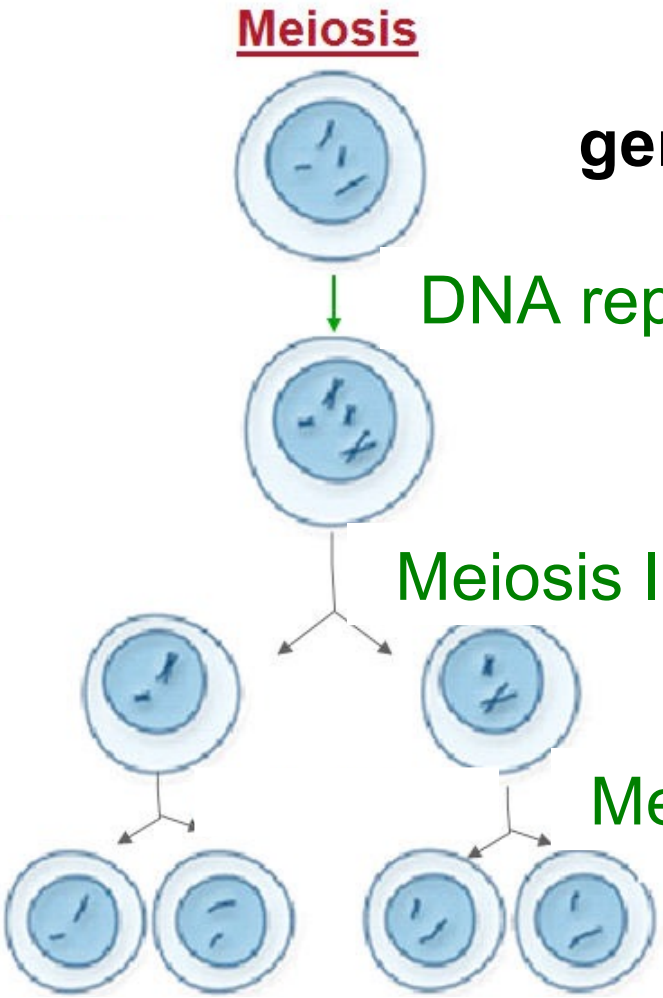




5 Key Ideas about the **inheritance** of cancer-causing mutations

1. **Mitosis occurs in somatic cells and produces two identical cells**
2. **Meiosis occurs in germline cells and produces four different gametes**
3. **Mutations primarily occur in DNA replication and repair in dividing cells (cells undergoing mitosis or meiosis)**
4. **New mutations can occur in somatic cells OR germline cells.**
5. **Only mutations in germline cells and gametes can be inherited.**

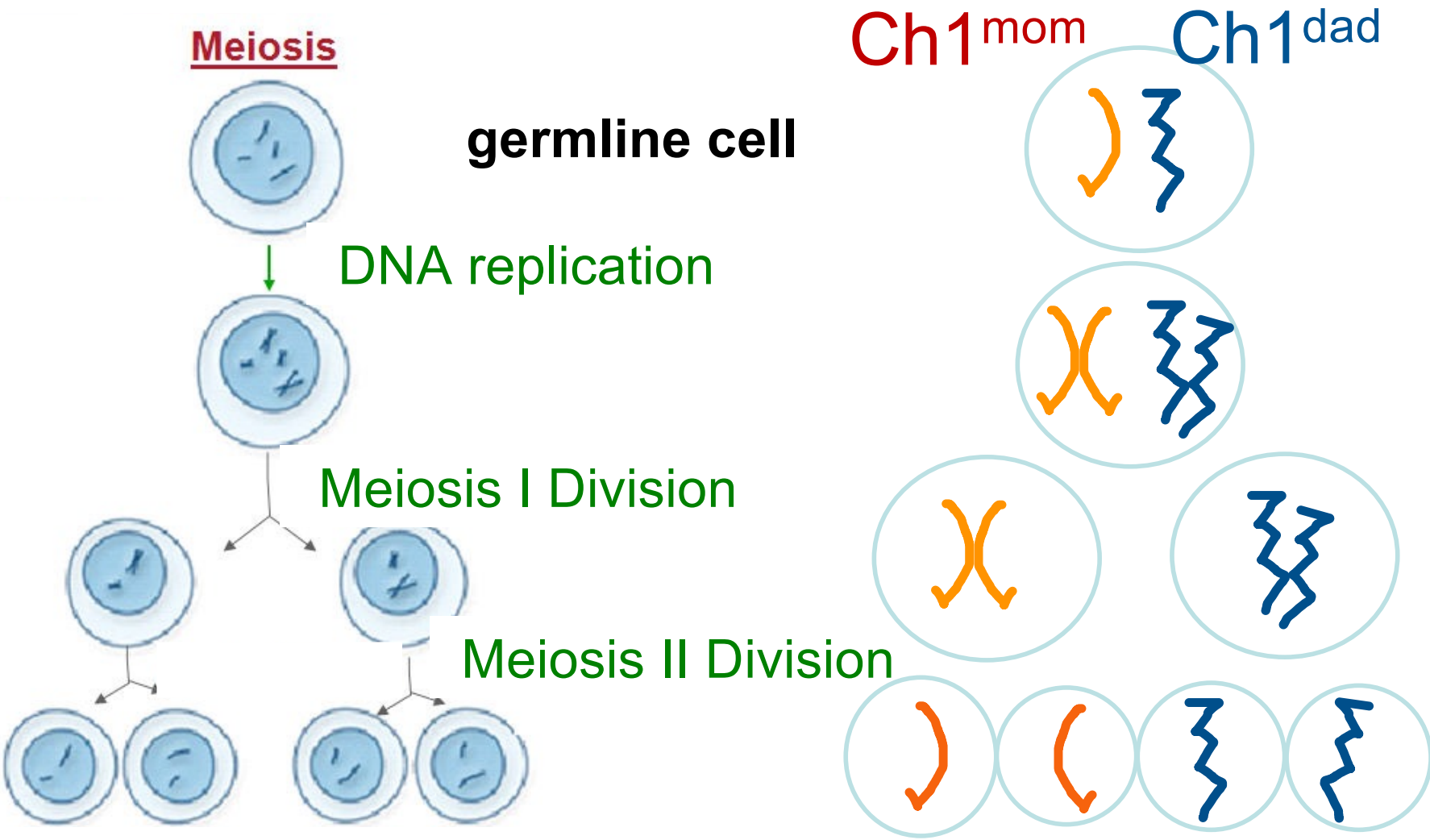
Key Idea 2: Meiosis occurs when germline cells divide to form four different gametes



Germline = All cells that will develop into gametes in your gonads (including the egg and sperm)

gametes (sperm cells in biological males, egg cells in biological females)

Key Idea 2: Meiosis occurs when germline cells divide to form four different gametes



gametes (sperm cells in biological males, egg cells in biological females)



Name Tent Reflection

Please THINK and WRITE (in a complete sentence) for today…

Please share one question you have about what we discussed in today's class…

_____

_____

Your Preferred First Name

Please leave all name tents in the bin at your table after EVERY class session!

Reflective Journal #6 preview

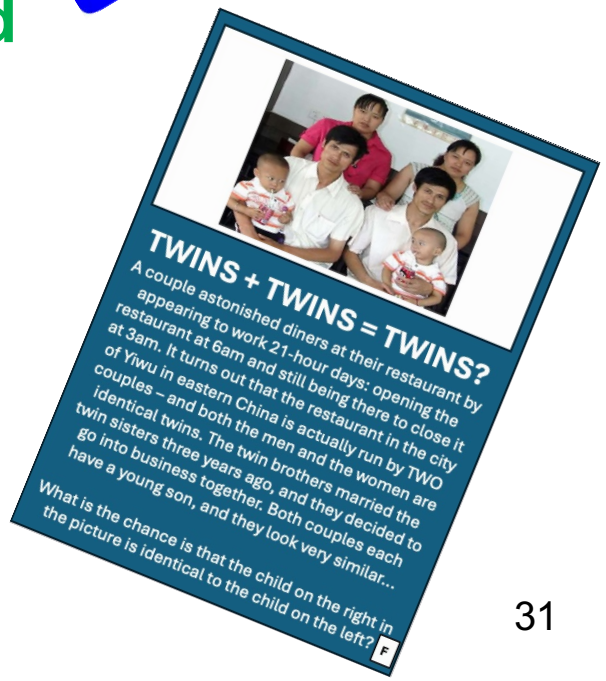
Due Sunday, September 27, 2026
by Midnight (11:59pm)

- **Complete Reflective Journal #6 Herman and Marisa AND Twins+Twins=Twins?**

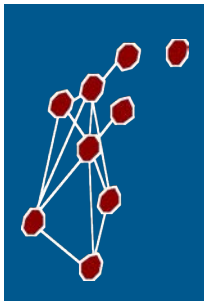
Cancer risk can differ among populations because people experience different environments. With Herman and Marisa, we're asking whether cancer risk can also run in families because people inherit different versions of genes.

Do identical twins make genetically identical gametes?








Assignments

Due Sunday, September 27, 2026
by Midnight (11:59pm)

- **Complete Reflective Journal #6 Herman and Marisa AND Twins+Twins=Twins?**

Due Saturday, September 26, 2026
by Midnight (11:59pm)

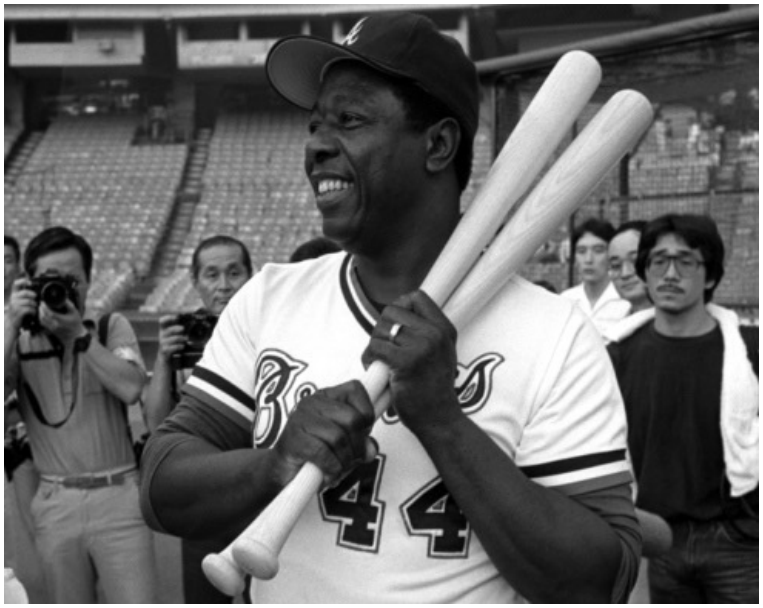
- **Self-Assessment Quiz #4: Mutations and Chance of Inheriting Cancer Mutations**

Opens Sunday, September 27, 2026
Due Saturday, October 3, 2026
by Midnight (11:59pm)

- **Self-Assessment Quiz #5: About Inheriting Cancer Mutations, Mitosis, and Meiosis**

Go Team Bio 100!

What is your prediction about how you scored on Exam #1…

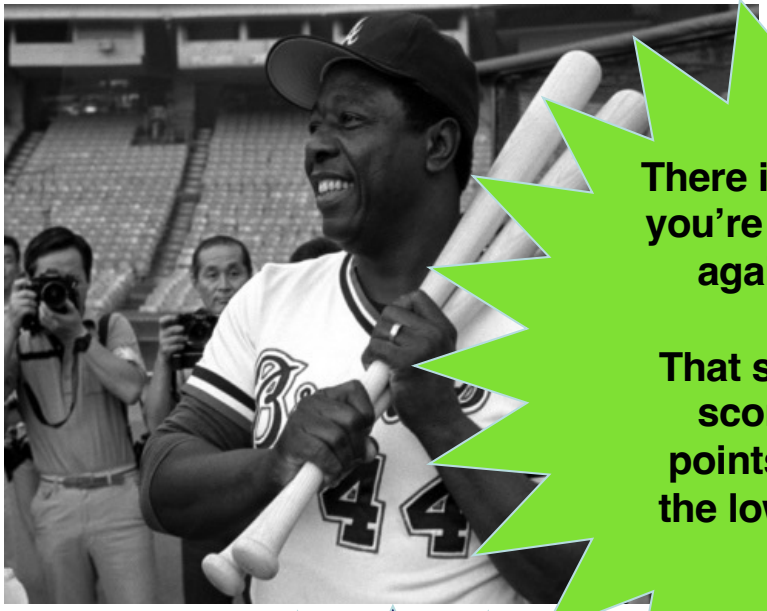


Maybe you're feeling like Hank Aaron "The Home Run King"



Maybe you didn't hit it out of the park THIS TIME…

What is your prediction about how you scored on Exam #1…



There is NO curve, and you're only competing against yourself!

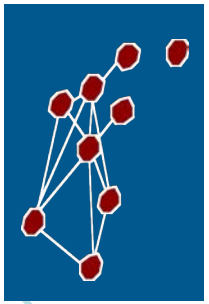
That said, the highest score (out of 110 points) was 110 and the lowest score was 35.



The majority of the course is still ahead of us. Be persistent!

Please don't define yourself by a grade! Remember why you are at SFSU…




Please CAREFULLY review your current overall grade summary – handed back with your exam – to see where you can improve!

Updated Biology 100 Grade Summary, as of 9-23-2026

Section 6

SAMPLE

Name

Possible Points	Due Date	Class Activities, Reflective Journals, Self-Assessment Quizzees, and Extra Credit (40% of your grade)	
10	first week	More About You Survey	10
10	8/27/26	Reflective Journal #1: About You and Your Aspirations	8
10	8/30/26	Reflective Journal # 2: Evidence for the Altoids Scare & Exploring Size and Scale	10
10	9/6/26	Reflective Journal #3: Henrietta Lacks and HeLa Cells	4
10	9/13/26	Reflective Journal #4: Exploring Human Biology in the News	0
10	8/29/26	Self Assessment Quiz #1: Welcome, Syllabus, and Supports for Team Bio100	10
10	9/5/26	Self Assessment Quiz #2: Thinking about Structure and Function in Human Biology	9
10	9/12/26	Self Assessment Quiz #3: Understanding Cancer–From DNA/Genes to Traits/Proteins	10
30	first quarter	In-Class Activities, Discussions, and Clicker Questions (7 classes, 5 points per class)	30
	Anytime	Extra Credit: One-Time Office Hour Visit (10 points)	
110		**YOUR Points Subtotal**	91
		TOTAL Points AVAILABLE	110
		YOUR % for Class Activities, Reflective Journals, Self-Assessment Quizzees, and Extra Credit	**82.7%**

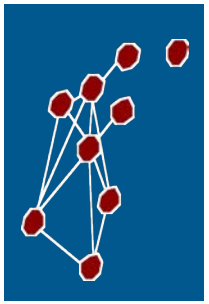
Possible Points	Due Date	Exams (60% of your grade)	
100	September 16,18	Exam #1	46
	October 12,13	Exam #2	
	November 4,5	Exam #3	
	December 16,17	Exam #4	
100		**YOUR Points Subtotal**	46
		TOTAL Points AVAILABLE	100.0
		YOUR % for Exams	**46.0%**

YOUR Overall Current Bio 100 Course Percentage **60.7%**

*This is 0.40*YOUR % Class Activities, Reflective Journals, Self-Assessment Quizzes, and Extra Credit + 0.60*YOUR % Exams*

Please email us so we can support you!
kdtanner@sfsu.edu,
pasion@sfsu.edu




What do I need to do to be successful in Biology 100?

Use evidence

Ask questions

Be skeptical

Cultivate wonder

Identify confusions

Think like a biologist!

It's about FIGURING THINGS OUT (not just memorizing)

U ABC IT!



Go Team Bio 100!